

Exact Spectral Inheritance in Prime-Power Resolution Towers

A Splitting Theorem for Affine-Cocycle Transfer Operators,
with Application to the Collatz Carry Sum

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Abstract

For a prime power $q = p^e$, the accelerated Collatz carry sum generates a multiplicative walk on $K_e = \langle 2, 3 \rangle \leq (\mathbb{Z}/q\mathbb{Z})^\times$ and an associated family of phase-twisted transfer operators $L_{e,t,x}$, whose dominant twisted eigenvalue controls the rate of equidistribution of carry-sum residues at fixed p [1]. This note settles how this spectral picture behaves as the resolution e is refined, by proving an exact structural theorem for the whole tower $(L_e)_{e \geq 1}$. We show that the natural pullback $J_e : H_e \rightarrow H_{e+1}$ along the reduction $\pi_e : K_{e+1} \rightarrow K_e$ not only intertwines the operators ($L_{e+1}J_e = J_eL_e$, under a single hypothesis on the multiplicative order of 2), but that its image is a *reducing*, not merely invariant, subspace: the orthogonal complement Z_{e+1} (functions of zero sum on every fiber of π_e) is also exactly invariant, giving an orthogonal direct-sum splitting $L_{e+1} \cong L_e \oplus N_{e+1}$. This is strictly stronger than the block-upper-triangular structure a bare invariant subspace would give, and it has an immediate consequence: the spectral-gap quantity $g_0(p, e)$ governing equidistribution is non-increasing in e at every step where the hypothesis holds. By the lifting-the-exponent lemma the hypothesis holds at every step $e \geq w_p(2)$, where $w_p(2) = v_p(2^{p-1} - 1)$, so $g_0(p, \cdot)$ is eventually monotone and convergent for every prime $p \nmid 6$, and monotone from $e = 1$ for every non-Wieferich prime. This resolves, in the monotone-and-convergent direction, an open question raised by preliminary numerical data that had suggested an ongoing decay. We verify that the hypothesis fails at a concrete prime (the Wieferich prime 1093, base 2, at the single step $e = 1 \rightarrow 2$), and explain why a different anomaly (the base-3 Wieferich prime 11) perturbs the spectral data without touching the hypothesis. The theorem and its proof are stated for a general affine-cocycle operator family; the Collatz carry sum is the motivating instance, not a hypothesis.

Keywords: transfer operators; spectral gap; prime-power towers; reducing subspaces; Wieferich primes; accelerated Collatz map; equidistribution.

MSC 2020: 11B83 (primary); 37A30, 47A15, 11A07 (secondary).

Scope of claims. *Proved here, generically* (Theorems 3, 6, 9): the intertwining identity; the exact orthogonal splitting; monotonicity and convergence of $g_0(p, e)$ at every step satisfying (H1). *Proved here, number-theoretically* (Propositions 13, 14): the equivalence of the Fermat-quotient and order-stall forms of the Wieferich condition, and the exact set of steps at which (H1) holds. *Verified computationally* (exact/high-precision, small cases): the splitting identity and the eigenvalue-union formula, for several primes and a non-Collatz control multiplier. *Explicitly open:* whether a non-primitive stratum's new-sector operator can ever exceed its own inherited eigenvalue (Open Problem 12); whether $\lim_e g_0(p, e)$ is strictly positive. *Not claimed:* anything about Collatz cycle closure, divergence, or the Collatz conjecture. This note is purely about the statistical/spectral structure of the carry-sum residue system.

1 Introduction

The $3x + 1$ problem and its accelerated (Syracuse) form have a long literature on the statistical behaviour of iterates and their residues; see Lagarias [7, 8] for surveys and Tao [10] for the strongest current results on equidistribution of Syracuse random variables modulo powers of 3.

Fixed-Prime Equidistribution of Accelerated Collatz Carry Sums [1] proves that, for the accelerated Collatz carry sum $C_L(D) = \sum_{j < L} 3^{L-1-j} 2^{s_j}$ over valuation words D , the residue $C_L(D) \bmod p$ equidistributes as $L \rightarrow \infty$, for every fixed prime $p \nmid 6$, at an explicit geometric rate governed by a transfer-operator spectral gap $g_0(p)$ (transfer operators in this sense are the standard tool of thermodynamic formalism; see Baladi [4]). The companion instrument **amgap** [2] extends the construction to prime powers $q = p^e$ and reports two data points suggesting the gap might shrink with e : $g_0(11, 1) = 0.3340$, $g_0(11, 2) = 0.3225$, explicitly flagging whether $g_0(p, e) \rightarrow 0$ as $e \rightarrow \infty$ as open.

A preliminary numerical extension (unpublished working notes) computed $g_0(11, 3) \approx 0.305$, apparently continuing the downward trend, and an initial structural analysis of the prime-power tower proposed a recursion $B_p(e) = \max\{P_p(e), B_p(e-1)\}$ (with $P_p(e)$ the dominant primitive-stratum eigenvalue and $B_p(e)$ the overall gap-defining quantity), based on an embedding of lower-level spectra into higher levels. On audit, that recursion was found to rest on an unproved assumption, namely that an embedded (inherited) eigenvalue remains globally dominant within its own stratum, and the monotonicity conclusion was downgraded to conditional.

This note resolves the question. The missing structural fact is that the pullback along the reduction map is not merely an invariant-subspace embedding but splits the operator as an honest orthogonal direct sum. This upgrades the available inequality to an exact formula, from which monotonicity follows by a short, elementary argument (Theorem 9). We also locate precisely the one arithmetic hypothesis the construction needs (Section 6), determine exactly at which levels it holds for every prime, and exhibit a concrete prime where it fails.

In the language of the Closure–Separation–Stability audit protocol for reduced representations [3], the reduction π_e is a coarse-graining of the residue state space. Theorem 6 says that this coarse-graining is exact in a strong sense: the coarse observables close under the dynamics (the level- e operator is recovered exactly), and the information discarded by the reduction separates into its own invariant sector rather than leaking back into the coarse one.

2 The operator family

Definition 1 (Affine-cocycle transfer operator). Fix an odd prime p and coprime integers $a, b \geq 2$ with $p \nmid ab$. For $e \geq 1$, let $q = p^e$ and $K_e = \langle a, b \rangle \leq (\mathbb{Z}/q\mathbb{Z})^\times$, a cyclic group (every subgroup of $(\mathbb{Z}/p^e\mathbb{Z})^\times$ is cyclic for odd p [6]). Let $r_e = \text{ord}_q(b)$. For $t \in \mathbb{Z}/q\mathbb{Z}$ and x in the open unit disk, define $L_{e,t,x} : H_e \rightarrow H_e$ on $H_e := \mathbb{C}^{K_e}$ by

$$(L_{e,t,x}f)(u') = \sum_{u \in K_e} e_q(tu) \frac{x^{d_0(u,u')}}{1 - x^{r_e}} f(u),$$

where $e_q(y) = \exp(2\pi i y/q)$ and $d_0(u, u') \in [1, r_e]$ is the least d with $u' \equiv u b^d a^{-1} \pmod{q}$ (the entry is 0 if no such d exists).

The Collatz instance is $(a, b) = (3, 2)$, with $K_e = \langle 2, 3 \rangle$ the reachable states of the multiplicative walk $u_{j+1} = u_j b^{D_j} a^{-1}$ underlying $C_L(D) \bmod q$ [1]. The relevant evaluation point is the saddle $x_* = 1 - 1/\log_2 b$ (for $(a, b) = (3, 2)$: $x_* = 1 - 1/\log_2 3$). At $t = 0$ every column of the matrix sums to $\sum_{d=1}^{r_e} x^d / (1 - x^{r_e}) = x / (1 - x)$, so $\rho(L_{e,0,x_*}) = \log_2 b - 1$ exactly, independent of p and e [1].

We fix the normalized inner product $\langle f, g \rangle_e := \frac{1}{|K_e|} \sum_{u \in K_e} f(u) \overline{g(u)}$ throughout (uniform probability measure on K_e), the normalization making the resolution maps below isometric.

Definition 2 (Resolution maps). $\pi_e : K_{e+1} \rightarrow K_e$ is reduction mod p^e , restricted to K_{e+1} ; it is a surjective group homomorphism with constant fiber size $m := |K_{e+1}|/|K_e| \in \{1, p\}$. Define

$$\begin{aligned} J_e : H_e &\rightarrow H_{e+1}, & J_e f &:= f \circ \pi_e & (\text{pullback}), \\ S_e : H_{e+1} &\rightarrow H_e, & (S_e f)(\bar{u}) &:= \sum_{u \in \pi_e^{-1}(\bar{u})} f(u) & (\text{fiber sum}). \end{aligned}$$

A direct computation shows that J_e is injective and, with the normalization above, an exact isometry, and that $S_e \circ J_e = m \cdot \text{id}$.

3 Intertwining and the splitting theorem

Theorem 3 (Pullback intertwining). *Suppose (H1): $r_{e+1} = p r_e$. Then for every $t \in \mathbb{Z}/p^e \mathbb{Z}$,*

$$L_{e+1,pt,x} J_e = J_e L_{e,t,x}.$$

Proof. Under (H1) the subgroup $\langle b \rangle$ grows by a factor p , so $m = p$. Fix a target $u' \in K_{e+1}$ and a source class $\bar{u} \in K_e$. A source $u \in \pi_e^{-1}(\bar{u})$ is joined to u' by an increment $d \in [1, r_{e+1}]$ exactly when $u = u' a b^{-d}$, and $u' a b^{-d}$ lies over $\bar{u}$ exactly when $d \equiv d_0(\bar{u}, \pi_e u') \pmod{r_e}$. Hence the admissible increments are $d_0 + k r_e$, $k = 0, \dots, p-1$ (here (H1) guarantees that these are exactly $p = m$ values in $[1, r_{e+1}]$, giving p distinct sources, i.e. the whole fiber). The phase $e_{p^{e+1}}(ptu) = e_{p^e}(t\bar{u})$ is constant on the fiber. Summing,

$$\sum_{u \in \pi_e^{-1}(\bar{u})} (M_{pt}(x))_{u',u} = e_{p^e}(t\bar{u}) \sum_{k=0}^{p-1} \frac{x^{d_0 + k r_e}}{1 - x^{r_{e+1}}} = e_{p^e}(t\bar{u}) \frac{x^{d_0}(1 - x^{p r_e})}{(1 - x^{r_e})(1 - x^{r_{e+1}})} = e_{p^e}(t\bar{u}) \frac{x^{d_0}}{1 - x^{r_e}},$$

using $r_{e+1} = p r_e$ in the last step; this is exactly $(M_t(x))_{\pi_e u', \bar{u}}$. Applying both sides to a function constant on fibers gives the claim. $\square$

Remark 4. The proof never uses $\gcd(t, p) = 1$; it lifts any residue t at the lower level by one factor of p , not only primitive ones.

Lemma 5 (Pushforward intertwining). *Under (H1), $S_e \circ L_{e+1,pt,x} = L_{e,t,x} \circ S_e$.*

Proof. The same geometric-series computation, now with the source $u \in K_{e+1}$ fixed and the sum taken over targets $u' \in \pi_e^{-1}(\bar{u}')$, gives $\sum_{u' \in \pi_e^{-1}(\bar{u}')} (M_{pt})_{u',u} = (M_t)_{\bar{u}', \pi_e u}$. Therefore

$$(S_e L_{e+1} f)(\bar{u}') = \sum_u f(u) \sum_{u' \in \pi_e^{-1}(\bar{u}')} (M_{pt})_{u',u} = \sum_{\bar{u}} (M_t)_{\bar{u}', \bar{u}} \sum_{u \in \pi_e^{-1}(\bar{u})} f(u) = (L_e S_e f)(\bar{u}'). \quad \square$$

Theorem 6 (Splitting). *Under (H1), $Z_{e+1} := \ker S_e$ satisfies:*

1. $Z_{e+1} = J_e(H_e)^\perp$ exactly;
2. Z_{e+1} is invariant under $L_{e+1,pt,x}$;
3. $H_{e+1} = J_e(H_e) \oplus Z_{e+1}$ is an orthogonal direct sum of two $L_{e+1,pt,x}$ -invariant subspaces, and in an orthonormal basis respecting this sum,

$$L_{e+1,pt,x} \cong L_{e,t,x} \oplus N_{e+1,pt,x}, \quad N_{e+1,pt,x} := L_{e+1,pt,x}|_{Z_{e+1}}.$$

Proof. (1) $\langle h, J_e \bar{g} \rangle_{e+1} = 0$ for all $\bar{g}$ iff $\sum_{u \in \pi_e^{-1}(\bar{u})} h(u) = 0$ for every $\bar{u}$, i.e. $S_e h = 0$. (2) If $S_e f = 0$ then $S_e(L_{e+1}f) = L_e(S_e f) = 0$ by Lemma 5. (3) Immediate from (1), (2) and Theorem 3: two orthogonal subspaces, each invariant and summing to the whole space, give an exactly block-diagonal matrix in any adapted orthonormal basis. Since J_e is an isometry, the block on $J_e(H_e)$ is unitarily equivalent to $L_{e,t,x}$. $\square$

This is strictly stronger than the invariant-subspace statement alone: Theorem 3 by itself gives only block-upper-triangularity, for which the complementary block need not be invariant. The operators here are not normal, so the invariance of Z_{e+1} is genuinely additional information, supplied by Lemma 5 (equivalently, $J_e(H_e)$ is invariant under the adjoint as well). Theorem 6 was verified independently by exact symbolic computation for $p = 5, 13$ (Section 7), including for non-primitive source residues t .

Corollary 7. $\text{Spec}(L_{e+1,pt,x}) = \text{Spec}(L_{e,t,x}) \sqcup \text{Spec}(N_{e+1,pt,x})$ (multiset union, with full Jordan structure inherited), and

$$\rho(L_{e+1,pt,x}) = \max \{ \rho(L_{e,t,x}), \rho(N_{e+1,pt,x}) \} \quad \text{exactly.}$$

Corollary 8. The singular-value spectrum of $L_{e+1,pt,x}$ is exactly the union of the singular-value spectra of $L_{e,t,x}$ and $N_{e+1,pt,x}$, since the adjoint is block-diagonal in the same orthonormal basis.

4 Monotonicity

Write $B_p(e) := \max_{t \neq 0} \rho(L_{e,t,x_*})$ and $g_0(p, e) := 1 - B_p(e)/\lambda_0$, where $\lambda_0 = \rho(L_{e,0,x_*})$ is constant in e .

Theorem 9 (Monotonicity). Suppose (H1) holds at the step $e \rightarrow e+1$. Then $B_p(e+1) \geq B_p(e)$, hence $g_0(p, e+1) \leq g_0(p, e)$.

Proof. Let t_* achieve $B_p(e) = \rho(L_{e,t_*,x_*})$ (it exists, the set being finite; t_* need not be primitive). Since $t_* \not\equiv 0 \pmod{p^e}$, the lift pt_* is nonzero mod p^{e+1} . By Corollary 7, $\rho(L_{e+1,pt_*,x_*}) = \max\{B_p(e), \rho(N_{e+1,pt_*,x_*})\} \geq B_p(e)$. Since $B_p(e+1)$ is a maximum over all nonzero residues, including pt_* , we get $B_p(e+1) \geq B_p(e)$. $\square$

Corollary 10. On any range of levels where (H1) holds at every step, $g_0(p, \cdot)$ is a non-increasing, nonnegative sequence, hence convergent. Nonnegativity: $\rho(L_{e,t,x_*}) \leq \rho(L_{e,0,x_*}) = \lambda_0$ for every t , since $|M_t| = M_0$ entrywise. By Proposition 14 below, this range always contains all $e \geq w_p(b)$, so $\lim_{e \rightarrow \infty} g_0(p, e)$ exists for every admissible p .

Remark 11. This does not determine the limit. Two logically separate questions remain open: whether only finitely many levels e set a new record for B_p (equivalently, whether the limit is attained at a finite level), and whether the limit is strictly less than λ_0 (mixing survives) or equal to it ($g_0 \rightarrow 0$). Theorem 9 answers neither; it establishes only that the sequence is eventually monotone and therefore has a limit.

The proof of Theorem 9 isolates exactly what remains open:

Open Problem 12 (Stratum-local dominance). For a non-primitive t (writing $t = pt'$), is

$$\rho(N_{e+1,pt',x_*}) \leq \rho(L_{e,t',x_*}),$$

i.e. does the new sector ever exceed its own inherited value? This is not needed for Theorem 9, which requires only one safe witness (the lift of the historical maximizer), but it is the natural finer question about the internal structure of the tower, and it is not resolved here.

5 Explicit model for the new sector

$Z_{e+1} = \ker S_e$ is concretely the space of functions on K_{e+1} with zero sum on every fiber of π_e . Since K_{e+1} is cyclic, fixing a generator g gives a canonical character basis χ_k ($k = 0, \dots, |K_{e+1}| - 1$, via the discrete logarithm); $J_e(H_e)$ is spanned exactly by the χ_k trivial on $\ker \pi_e$, and Z_{e+1} by the remaining ones. When K_{e+1} is the full unit group (primes for which $\langle a, b \rangle$ has index 1), this coincides with the classical characters of exact conductor p^{e+1} ; when K_{e+1} is a proper subgroup, the identification with primitive Dirichlet characters is not automatic and is not claimed.

The untwisted operator $L_{e+1,0,x}$ is a convolution on the cyclic group K_{e+1} , so the χ_k diagonalize it; they do not diagonalize $L_{e+1,t,x}$ for $t \neq 0$ in general, and no closed form for the eigenvalues of $N_{e+1,t,x}$ at $t \neq 0$ was found. Under (H1), $\dim Z_{e+1} = (p-1)|K_e|$: a constant fraction $(p-1)/p$ of the total dimension at every step, neither vanishing nor dominating.

6 Arithmetic anomalies

Proposition 13. *For $p \nmid a$: $a^{p-1} \equiv 1 \pmod{p^2}$ if and only if $\text{ord}_{p^2}(a) = \text{ord}_p(a)$.*

Proof. $\text{ord}_{p^2}(a) \in \{\text{ord}_p(a), p \cdot \text{ord}_p(a)\}$ always (standard order lifting for prime powers [6]). Now $\text{ord}_p(a) \mid p-1$ by Fermat, while $p \cdot \text{ord}_p(a) \geq p > p-1$ cannot divide $p-1$. So $a^{p-1} \equiv 1 \pmod{p^2} \iff \text{ord}_{p^2}(a) \mid p-1 \iff \text{ord}_{p^2}(a) = \text{ord}_p(a)$. $\square$

The following standard consequence of the lifting-the-exponent lemma determines exactly at which steps (H1) holds.

Proposition 14 (Where (H1) holds). *Let p be an odd prime, $p \nmid b$, $r_1 = \text{ord}_p(b)$, and $w = w_p(b) := v_p(b^{r_1} - 1) \geq 1$ (equivalently $w = v_p(b^{p-1} - 1)$). Then*

$$\text{ord}_{p^e}(b) = \begin{cases} r_1, & e \leq w, \\ r_1 p^{e-w}, & e \geq w. \end{cases}$$

In particular (H1) holds at the step $e \rightarrow e+1$ if and only if $e \geq w$. For non-Wieferich primes to base b ($w = 1$) it holds at every step.

Proof. For odd p and $p \mid c-1$, the lifting-the-exponent lemma gives $v_p(c^n - 1) = v_p(c-1) + v_p(n)$. Since $b^n \equiv 1 \pmod{p}$ forces $r_1 \mid n$, apply this with $c = b^{r_1}$: $v_p(b^{r_1 n'} - 1) = w + v_p(n')$. Thus $b^{r_1 n'} \equiv 1 \pmod{p^e}$ iff $v_p(n') \geq e - w$, and the smallest such n' is $p^{\max(0, e-w)}$. The equality $v_p(b^{r_1} - 1) = v_p(b^{p-1} - 1)$ follows from the same formula, since $(p-1)/r_1$ is prime to p . $\square$

$p = 11$ (**Wieferich base 3**). $3^{10} \equiv 1 \pmod{121}$, verified directly; 11 is one of exactly two known base-3 Wieferich primes (11 and 1006003) [9, 5]. On the other hand $w_{11}(2) = 1$, so by Proposition 14 the order of 2 lifts generically ($\times 11$) at every level, and (H1) holds throughout. Theorems 3–9 therefore apply at every step for $p = 11$, and the previously observed three-level transient in $P_{11}(e)$ before freezing is consistent with, and now known to sit inside, a provably monotone sequence. The base-3 anomaly affects the internal structure of the primitive-stratum operator (there being no inherited part there to split against) and is plausibly the reason for the specific numerical values of $P_{11}(e)$, but it does not interact with the splitting mechanism at all.

$p = 1093$ (**Wieferich base 2**). $2^{1092} \equiv 1 \pmod{1093^2}$, verified directly; 1093 is the smallest of the two known base-2 Wieferich primes [11, 5]. Direct computation gives $\text{ord}_{1093}(2) = 364$ and $\text{ord}_{1093^2}(2) = 364$ (not 364×1093), so (H1) fails exactly at $e = 1 \rightarrow 2$; by Proposition 14 ($w = 2$) it holds at every subsequent step (indeed $\text{ord}_{1093^3}(2) = 364 \times 1093$), so $g_0(1093, e)$ is non-increasing for $e \geq 2$. Notably, the group K_{e+1} itself still grows generically ($|K_2|/|K_1| = 1093$ exactly, since 3 is not simultaneously anomalous and compensates), but this does not rescue the proof of Theorem 3, because the geometric-series telescoping is tied specifically to the lifting of $\text{ord}(2)$, not to the overall size of the group. This illustrates that (H1) is the correct, minimal, and non-redundant hypothesis.

7 Numerical verification

All verification in this note used exact symbolic computation (**sympy**) or high-precision numerical linear algebra on small cases, not broad numerical sweeps. Confirmed exactly: (i) $S_e M_{e+1} = M_e S_e$ and invariance of $\ker S_e$, for $p = 5, 13$ at $e = 1 \rightarrow 2$, for both primitive and non-primitive lifted residues; (ii) the exact block-diagonal structure (off-diagonal blocks $< 10^{-14}$ in a properly orthonormalized basis) and the resulting exact singular-value union (Corollary 8), matched to 10^{-8} across the six leading values; (iii) $\rho(L_{\text{hi}}) = \max\{\rho(L_{\text{lo}}), \rho(N)\}$ exactly, for both primitive and non-primitive source residues, $p = 5, 13$; (iv) the Wieferich computations of Section 6, by direct modular exponentiation; (v) genericity of the splitting mechanism for a non-Collatz control multiplier $(a, b) = (5, 2)$ at an arbitrary (non-saddle) evaluation point, confirming that the proof uses no Collatz-specific arithmetic. An independent re-implementation, prepared for this version, reproduced (i)–(iii) for $p = 5, 11, 13$ with $(a, b) = (3, 2)$ at x_* (including the non-primitive lift $t = 5$ at $e = 2 \rightarrow 3$ for $p = 5$) and (v) for $p = 7, 13$ with $(a, b) = (5, 2)$ at $x = 0.37$; all intertwining and off-diagonal residuals were below 2×10^{-14} and the spectral radii agreed to ten digits. The verification scripts accompany this deposit in the **src/** directory.

8 Conclusion

The prime-power resolution tower of accelerated-Collatz-carry-sum transfer operators splits exactly and orthogonally at every step satisfying one checkable arithmetic hypothesis, and that hypothesis holds at every step beyond the base-2 Wieferich depth $w_p(2)$ (from the first step for every non-Wieferich prime). This upgrades a previously conditional monotonicity claim for the equidistribution spectral gap $g_0(p, e)$ to a theorem, and sharpens the remaining open question to a precise, stratum-local statement (Open Problem 12) that is no longer entangled with the coarser question the research program originally posed. The theorem and its proof are generic to affine-cocycle transfer operators of this shape; the Collatz carry sum is the motivating instance, not a hypothesis of any result proved here.

AI Use Disclosure

AI language-model assistants, including Claude (Anthropic), were used in preparing this note: to help draft and edit the text, typeset it in L^AT_EX, review the proofs, suggest standard literature references, and independently re-implement the numerical checks of Section 7. All mathematical statements, proofs, computations and references were checked by the author, who takes full responsibility for the content. AI systems are not listed as authors.

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References

- [1] E. De Jesús, *Fixed-Prime Equidistribution of Accelerated Collatz Carry Sums: A Transfer-Operator Spectral Gap, with Two Open Uniformity Problems*, Zenodo, 2026. doi:10.5281/zenodo.20694734.
- [2] E. De Jesús, *An Additive–Multiplicative Spectral-Gap Analyzer for the Fixed-Prime Collatz Carry-Sum Transfer Operator: Instrument, Validation Gates, and Two Exploratory Findings*, technical note, Zenodo, June 2026.
- [3] E. De Jesús, *Closure, Separation, Stability: A Three-Level Audit Protocol for Reduced Representations*, Zenodo, 2026. doi:10.5281/zenodo.21855642.
- [4] V. Baladi, *Positive Transfer Operators and Decay of Correlations*, Advanced Series in Nonlinear Dynamics, vol. 16, World Scientific, Singapore, 2000.
- [5] F. G. Dorais and D. Klyve, A Wieferich prime search up to 6.7×10^{15} , *Journal of Integer Sequences* **14** (2011), Article 11.9.2.
- [6] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics, vol. 84, Springer, New York, 1990.
- [7] J. C. Lagarias, The $3x + 1$ problem and its generalizations, *American Mathematical Monthly* **92** (1985), no. 1, 3–23.
- [8] J. C. Lagarias (ed.), *The Ultimate Challenge: The $3x + 1$ Problem*, American Mathematical Society, Providence, RI, 2010.
- [9] P. L. Montgomery, New solutions of $a^{p-1} \equiv 1 \pmod{p^2}$, *Mathematics of Computation* **61** (1993), no. 203, 361–363.
- [10] T. Tao, Almost all orbits of the Collatz map attain almost bounded values, *Forum of Mathematics, Pi* **10** (2022), e12.
- [11] A. Wieferich, Zum letzten Fermat’schen Theorem, *Journal für die reine und angewandte Mathematik* **136** (1909), 293–302.